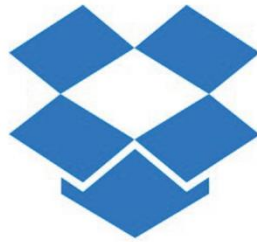


Cloud Technology Basics

Agenda

- Definition of cloud computing
- History and origins of cloud computing
- Cloud computing services and models
- Key cloud providers
- Other highly relevant topics
 - Utility-, distributed- and grid computing, Internet of Things, etc.

Cloud infrastructure everywhere!!

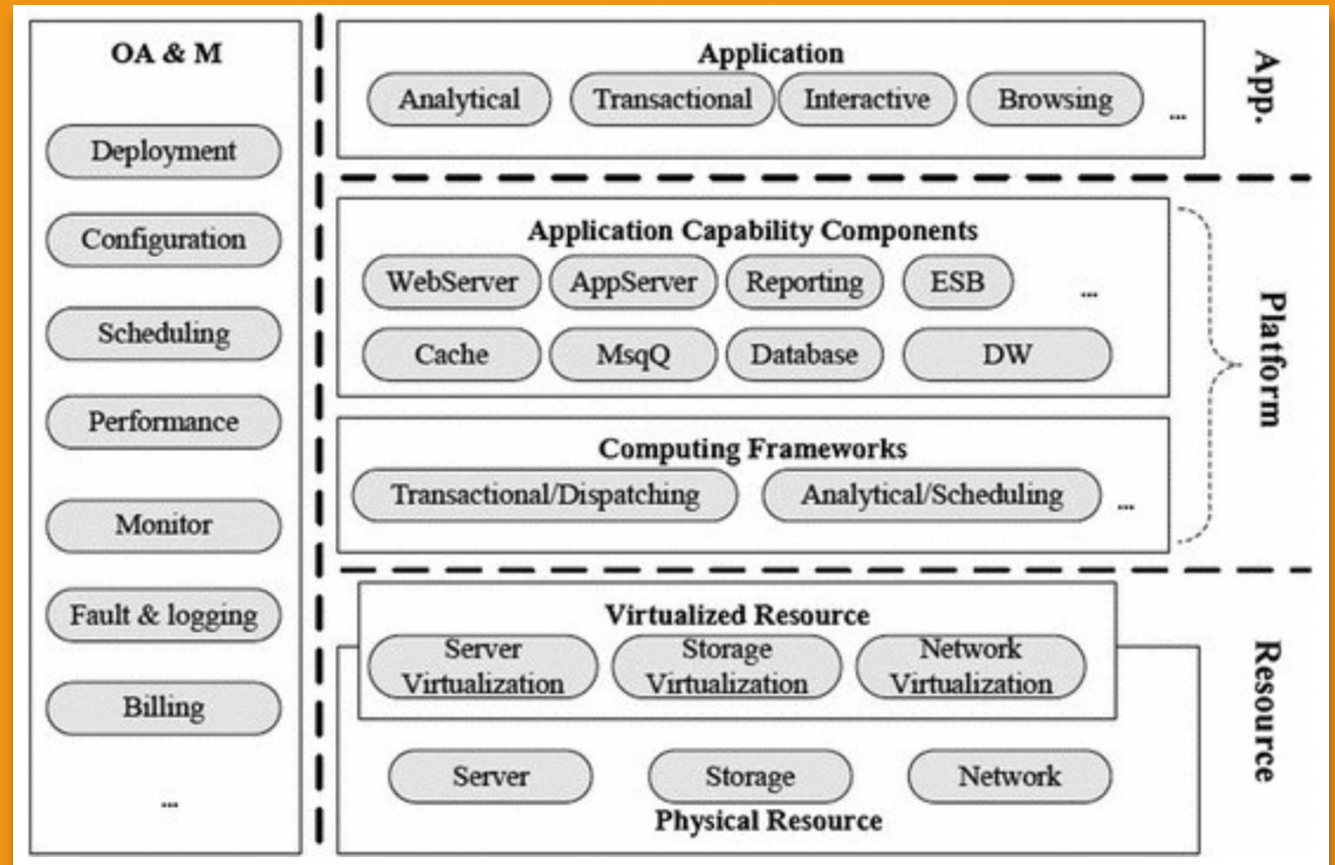


Definition

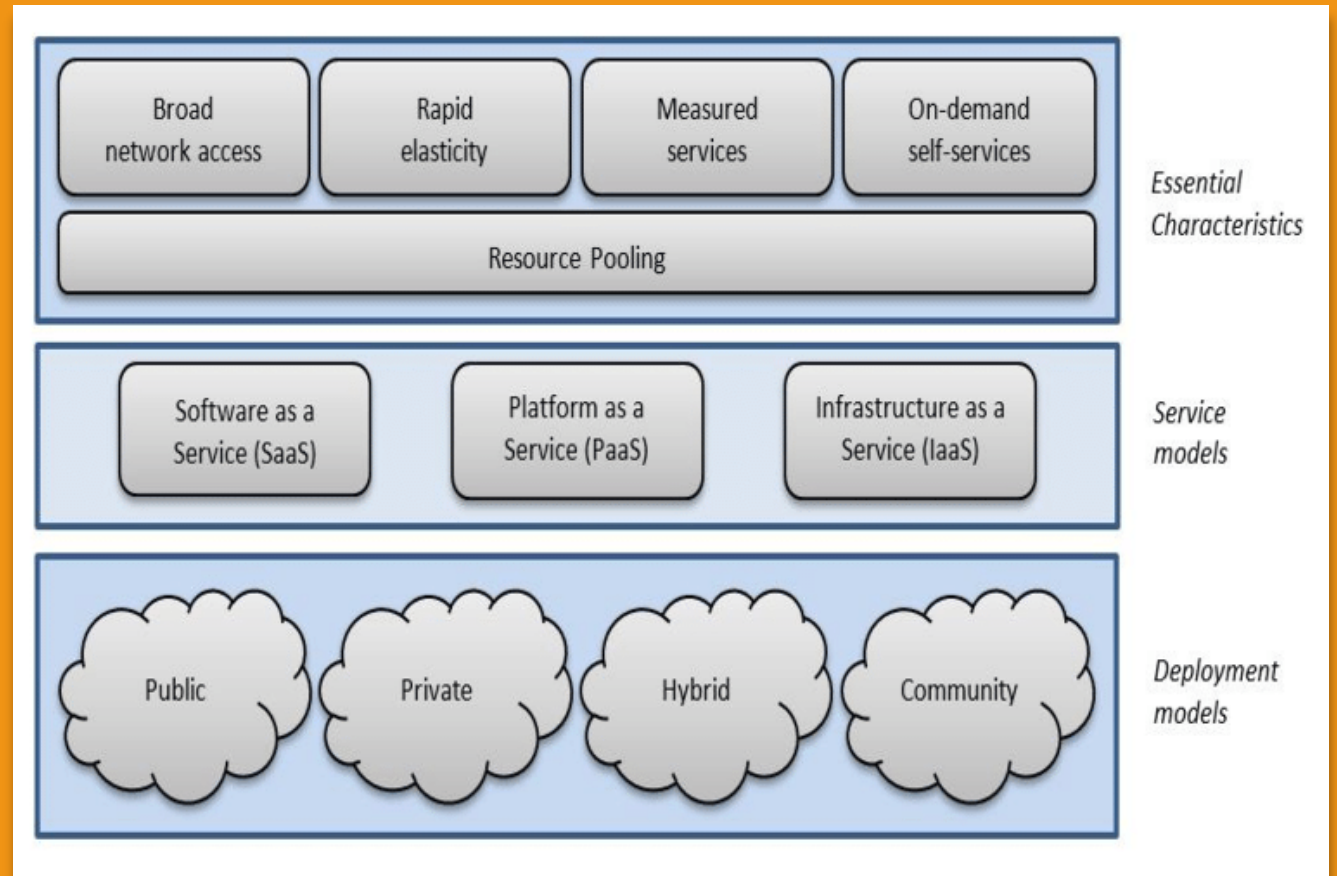
- *Computing and software resources that are delivered*
- *on demand, as service.*
 - *(2013, January) A Walk in the Clouds. Cloud Computing, CDW-G Reference Guide., 3-5.*
- *“The cloud” is one or multiple computers that store and transmit data.*
- *These servers are very large and can hold massive amounts of data.*
- *The servers can be housed anywhere in the world the user accesses the data by log-in from anywhere.*

Definition

- Using the internet to access someone else's software/hardware
- Large-scale distributed computer paradigm driven by
 - Economics of scale
 - Shared resource pool, abstraction, virtualization, scalability
 - Managed computing, storage, platforms and services



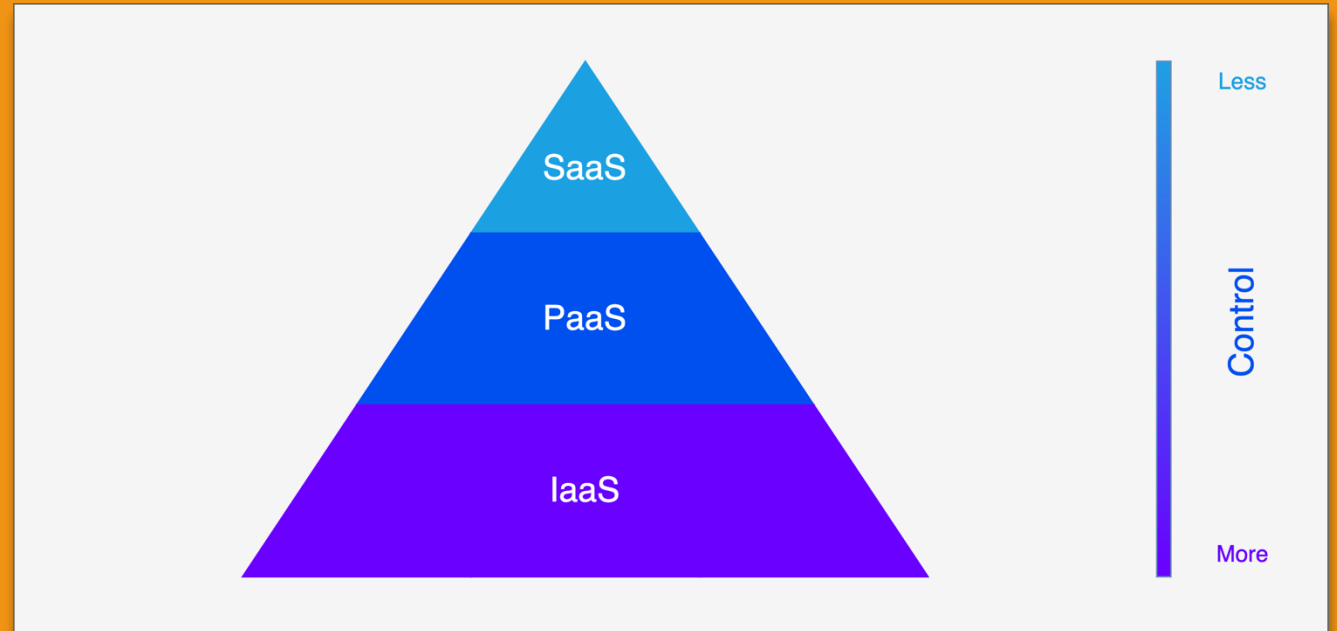
NIST Visual model of Cloud Computing



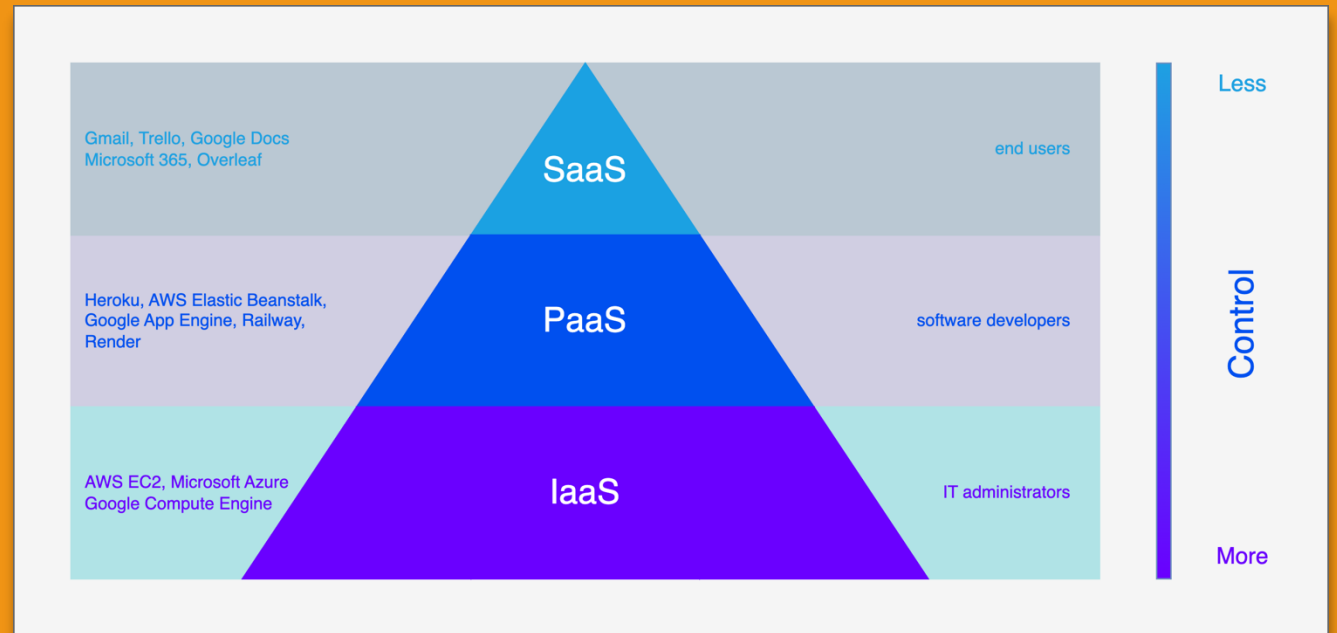
Essential characteristics

- On-demand self-services
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured services

Cloud Service Model



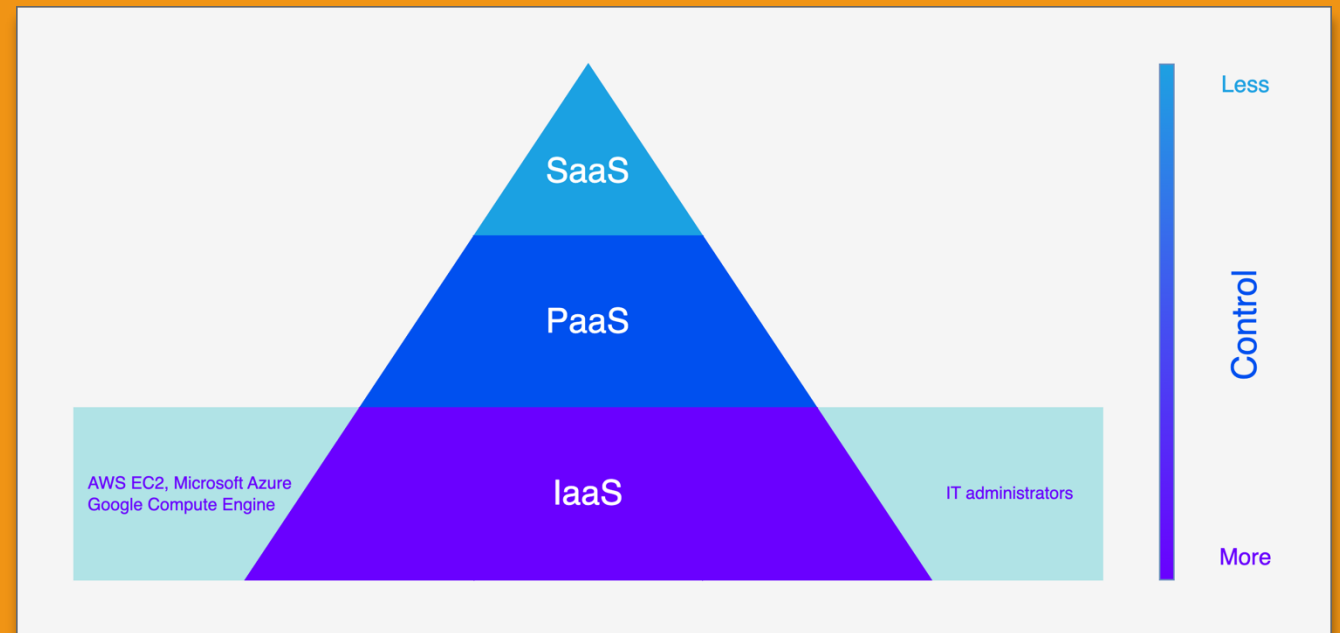
Cloud Service Model



Cloud Service Model

IaaS (Infrastructure as a Service)

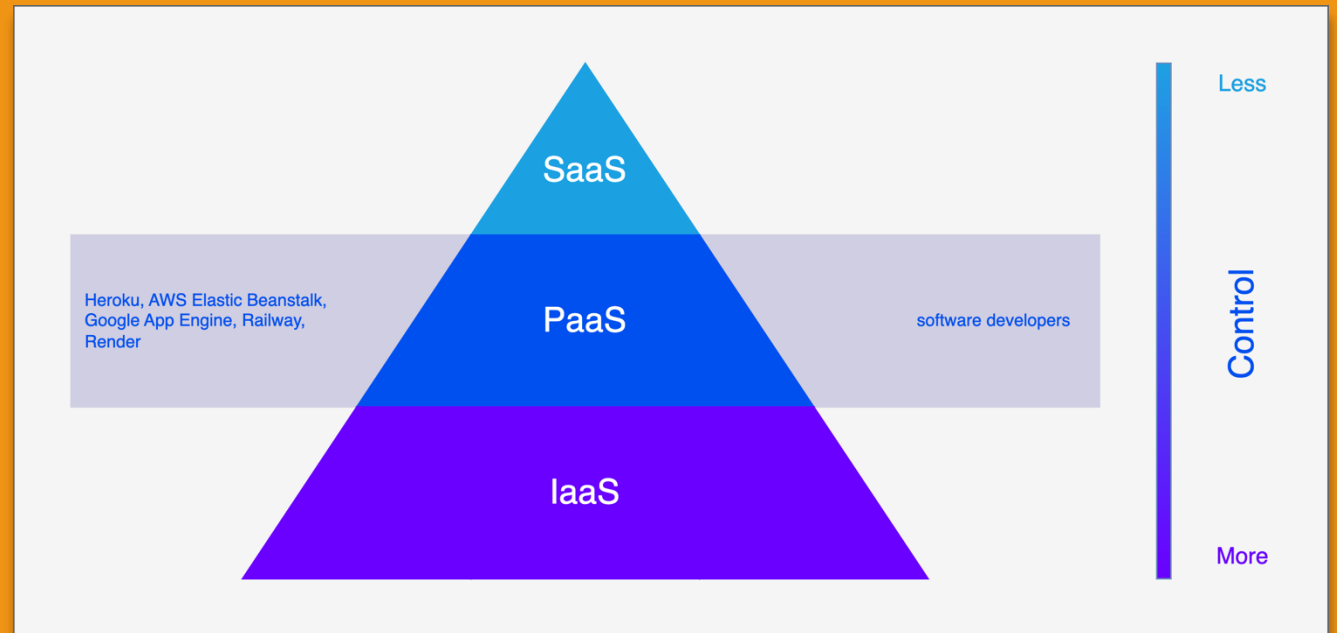
- IaaS is the delivery of technology infrastructure on demand
- Basically, you rent hardware or close-to-hardware
- Used by technical IT and server admins



Cloud Service Model

PaaS (Platform as a Service)

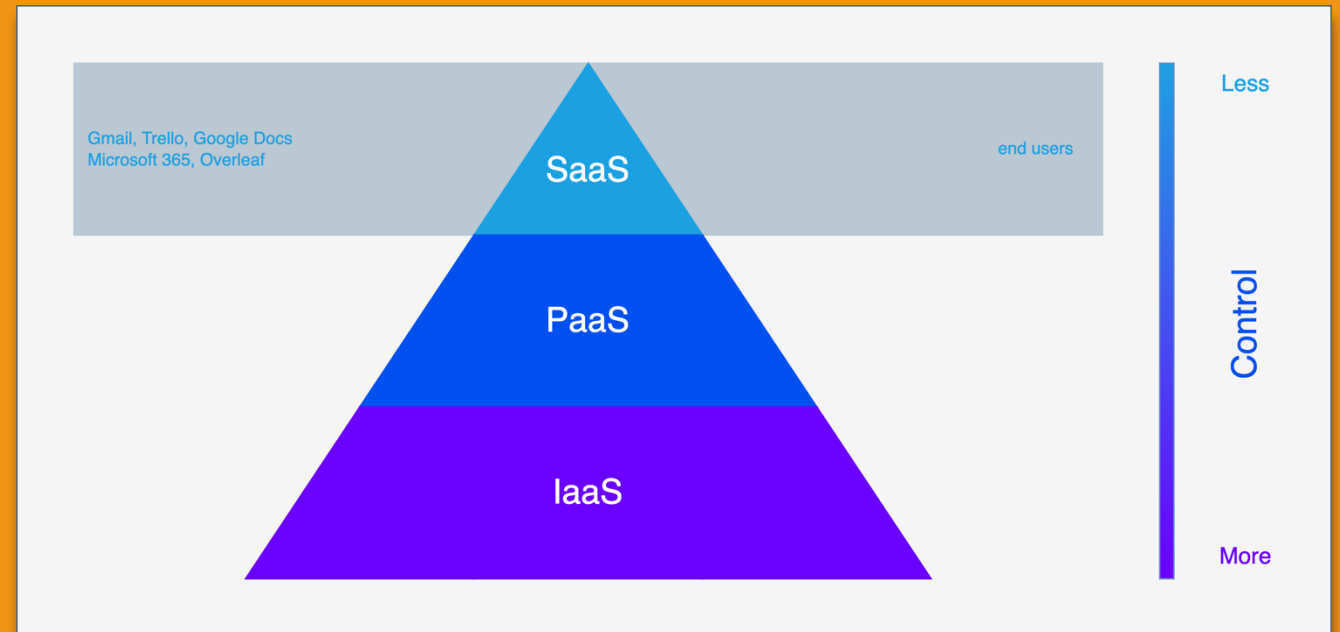
- PaaS provides all of the facilities required to support the complete life cycle of building and delivering web applications and services entirely from the internet.
- Basically, you rent a VM or environment services
- Used by developers
- ! Most relevant for our assignments !



Cloud Service Model

SaaS (Software as a Service)

- SaaS is a software delivery methodology that provides licensed multi-tenant access to software and its functions.
- Basically, online tools and products like Google Docs, Blackboard and Netflix
- Used by normies, business users, consumers, etc.

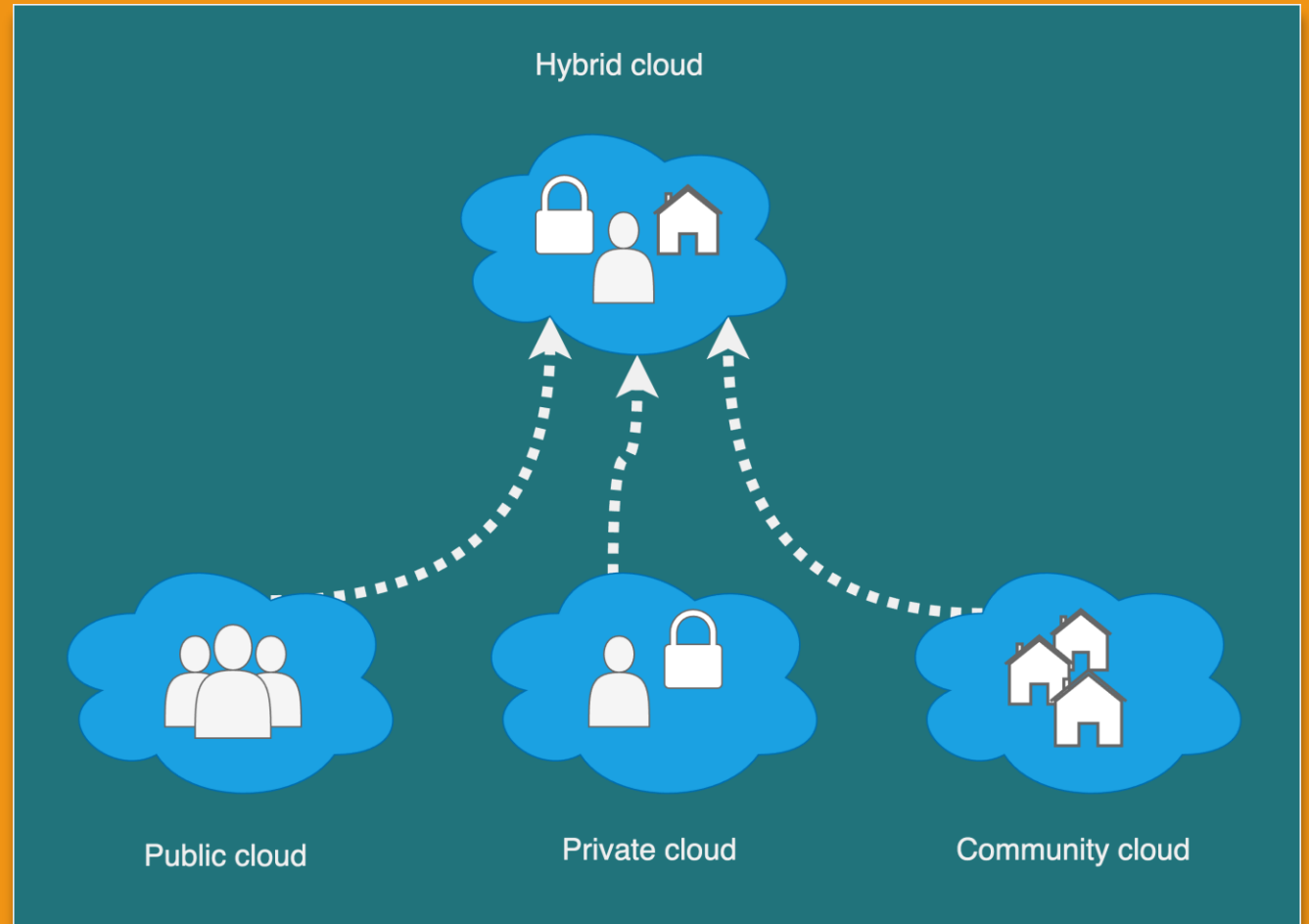


Cloud Service Model

SaaS Software as a Service	 BPM	 CRM	 DMS	 Social Portal	 Web Office
PaaS Platform as a Service	 Application Builder	 Mashup Studio	 Process Designer	 Report Editor	 Application Engines
IaaS Infrastructure as a Service	 Cloud Appliance	 Cloud Controller	 Elastic Compute	 Elastic Storage	 Elastic Database

Cloud Deployment Models

- **Public cloud**
 - A type of hosting which cloud services are delivered over a network for public use.
 - E.g., AWS, GCP, MS Azure
- **Private cloud**
 - Cloud infrastructure solely used by one organization.
 - E.g., NTNU's SkyHigh or a home server
- **Hybrid cloud**
 - Uses both private and public clouds but can remain separate entities.



Cloud Deployment Models

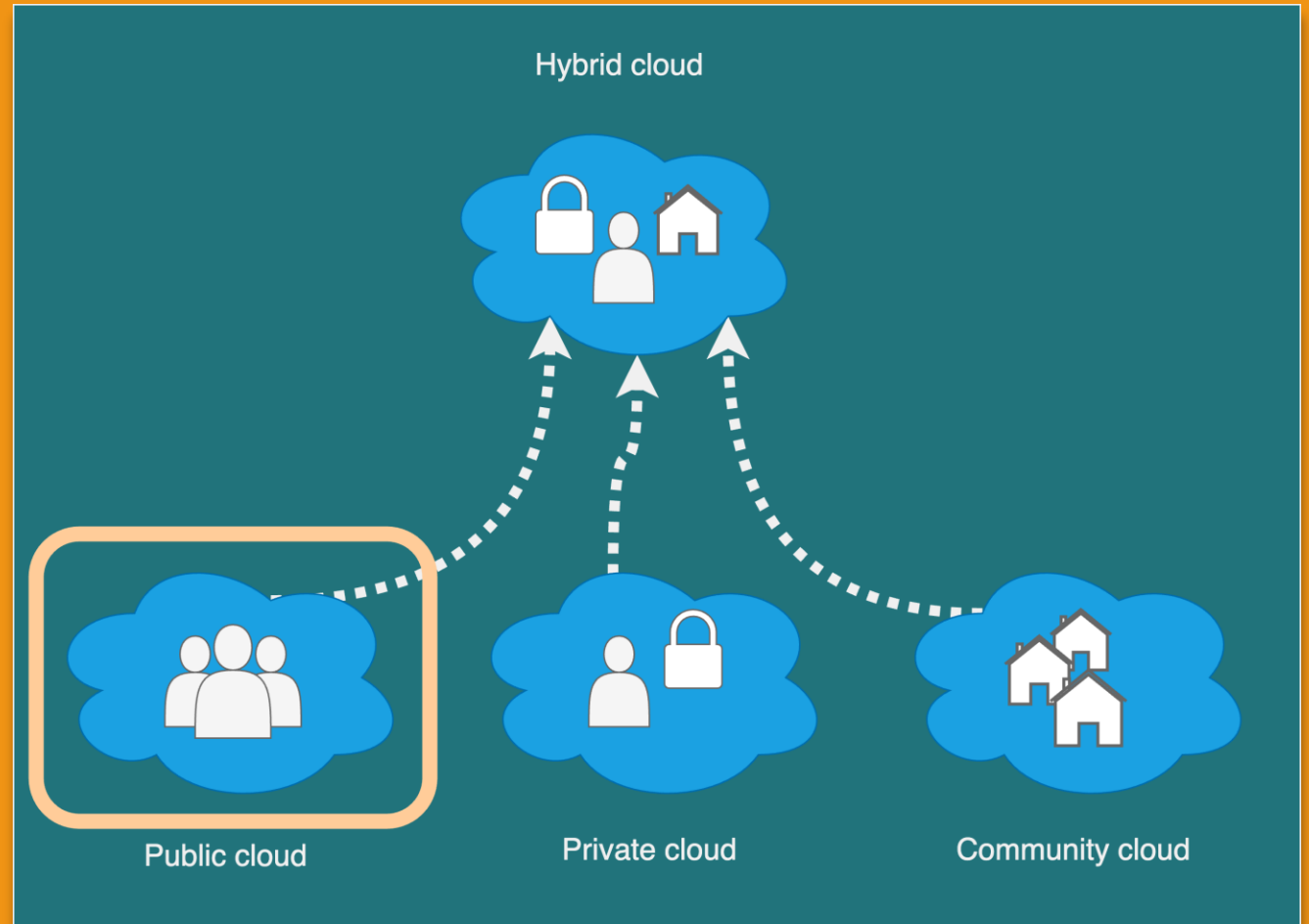
- Public clouds are owned by a company/organization, and you rent server space

Pros

- Cheap: No large up-front costs, lower economic risk
- Bigger resource pool, scalability
- Less management and responsibility

Cons

- Lack of control: Must trust Amazon, Google, etc.
- Can be more expensive in the long run



Cloud Deployment Models

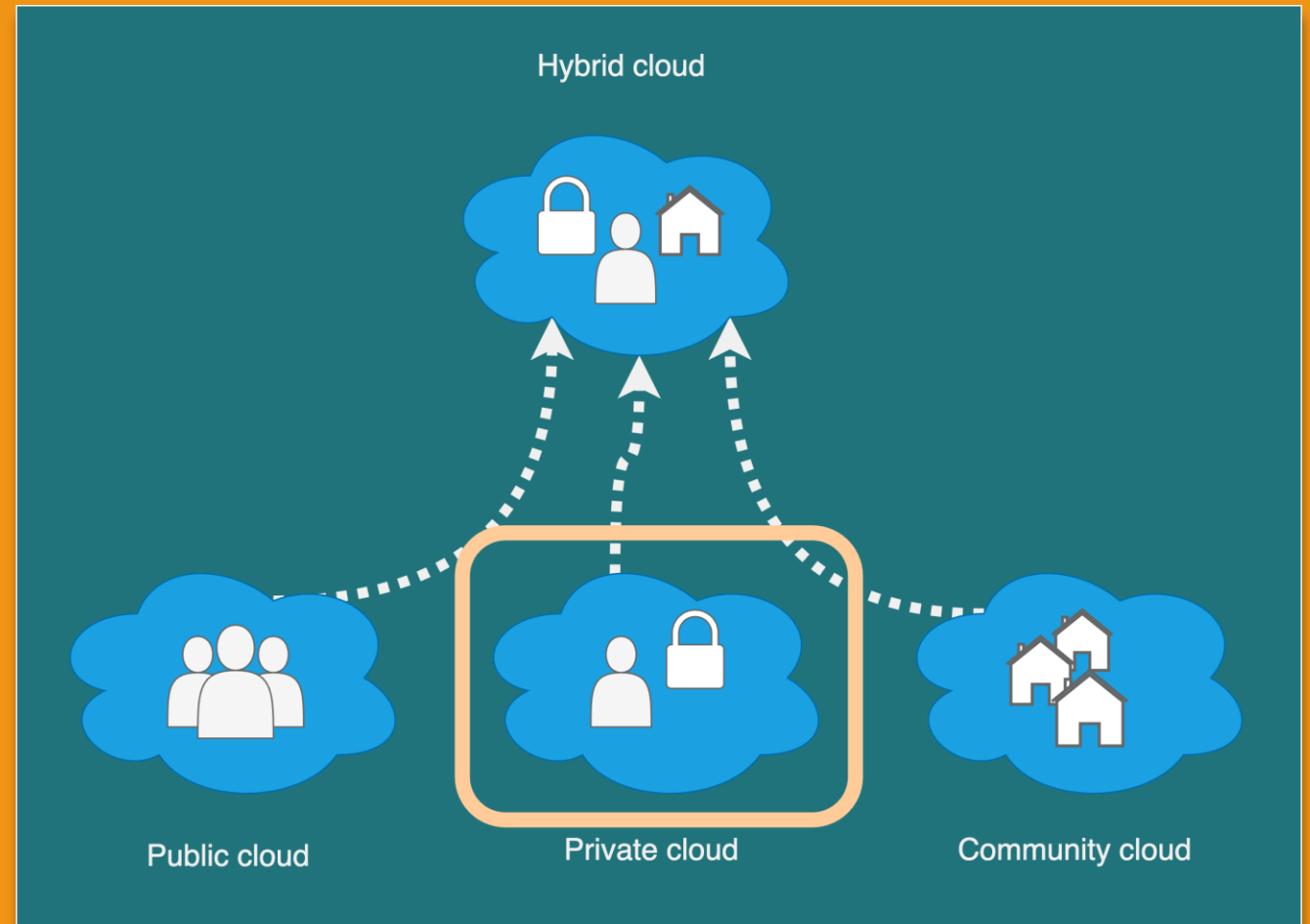
- Private clouds are your own servers, within your own networks

Pros

- More control and flexibility
- Better security: You don't have to trust anybody
- Can be cheaper in the long run

Cons

- High up-front costs
- Higher infrastructure complexity and more management



Cloud Deployment Models

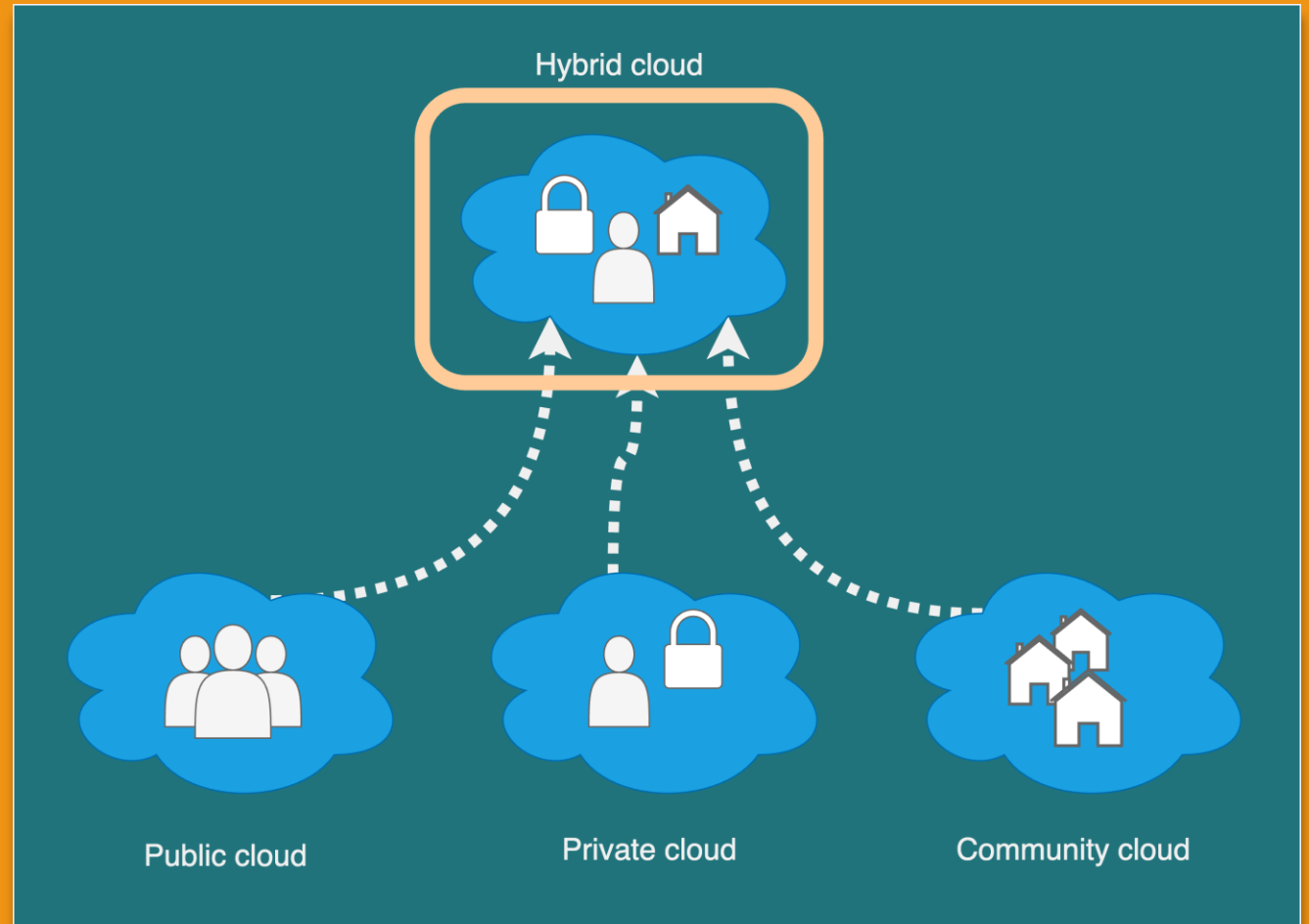
- Hybrid clouds are a combination between public and private

Pros

- The best of both worlds
- Security, flexibility + big resource pool where necessary

Cons

- Higher system complexity



Enterprise Cloud Solutions



Hybrid cloud

Scalability (public) +
security (private)



Disaster recovery

Backup images ready to
be re-instated quickly



Cloud file storage

Backup or archive
company data to cloud
storage



Load balancing

Cloud infrastructure for
overflow management
during peak usage

Enterprise Cloud Solutions



Overhead control

Lower overhead costs and make your bids more competitive



Distributed network control and cost reporting

Create an individual private networks for each of your subsidiaries or contracts



Rapid deployment

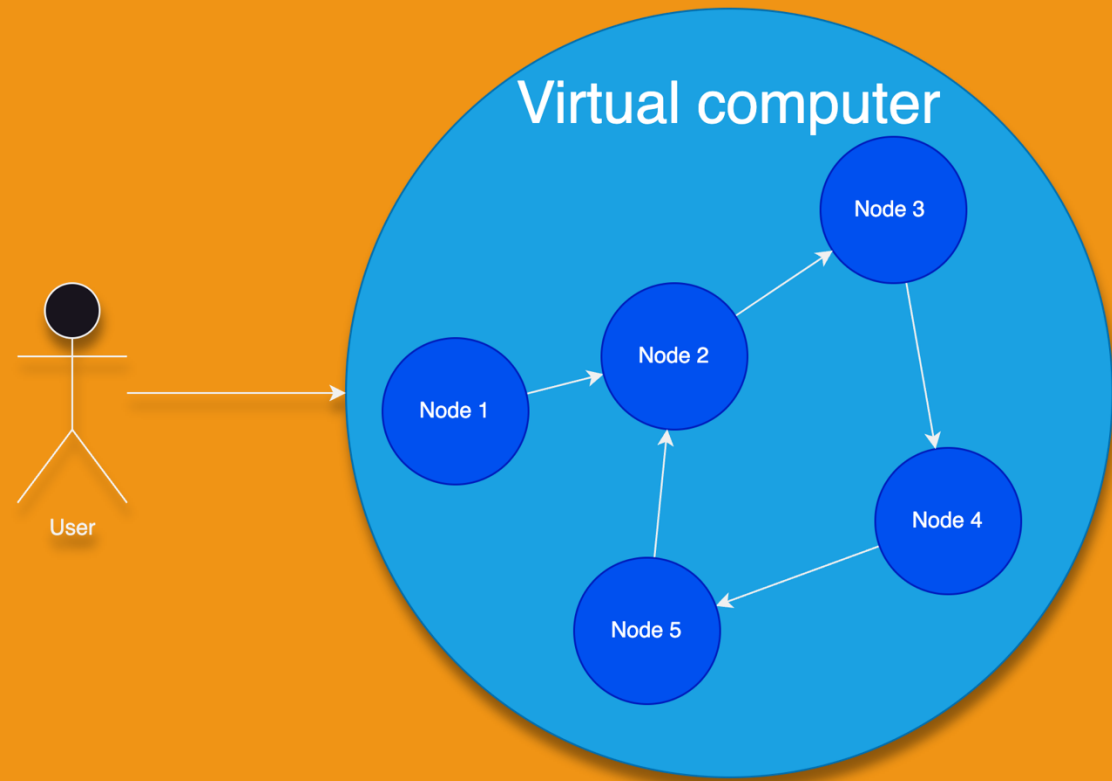
Turn up servers immediately to fulfil project timelines

Utility computing

- “Pay as you go” model
 - Replace capital costs with use-based charges

Distributed computing

Collection of autonomous computing elements which “looks like” a single system to its user



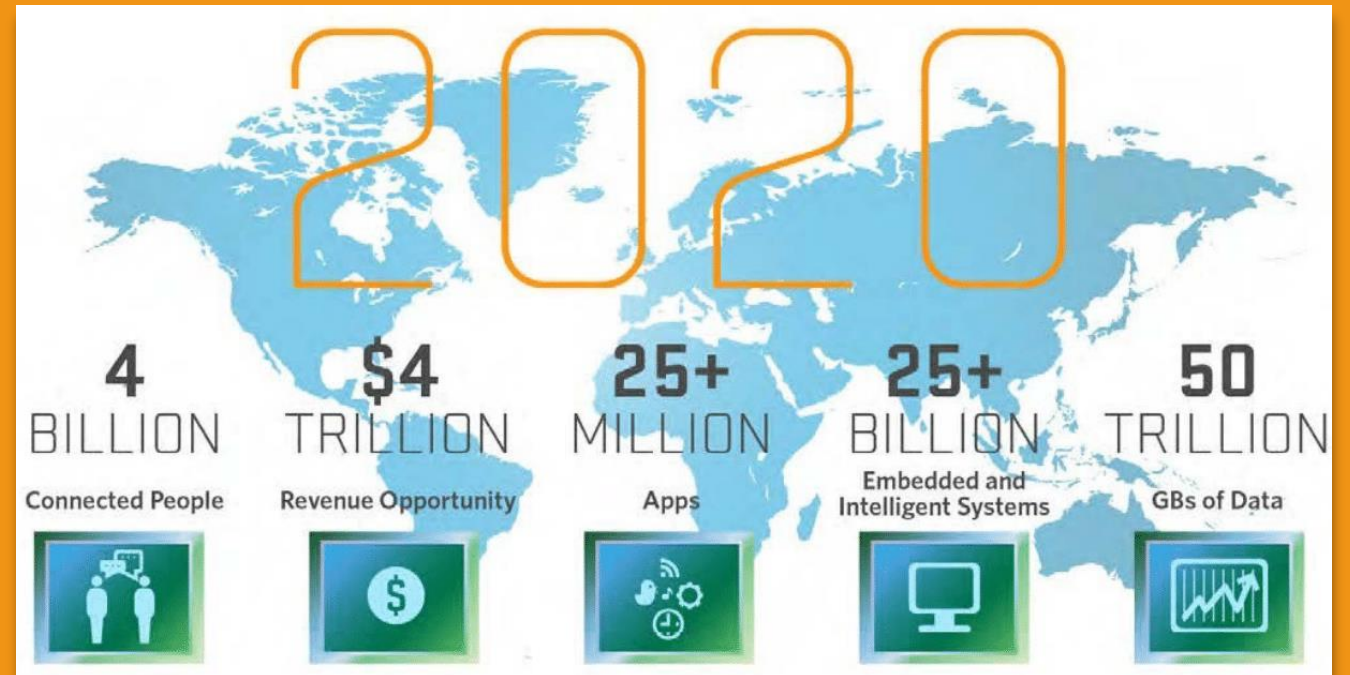
Distributed computing

- Support for resource sharing
- Distribution transparency
- Scalability
- Openness

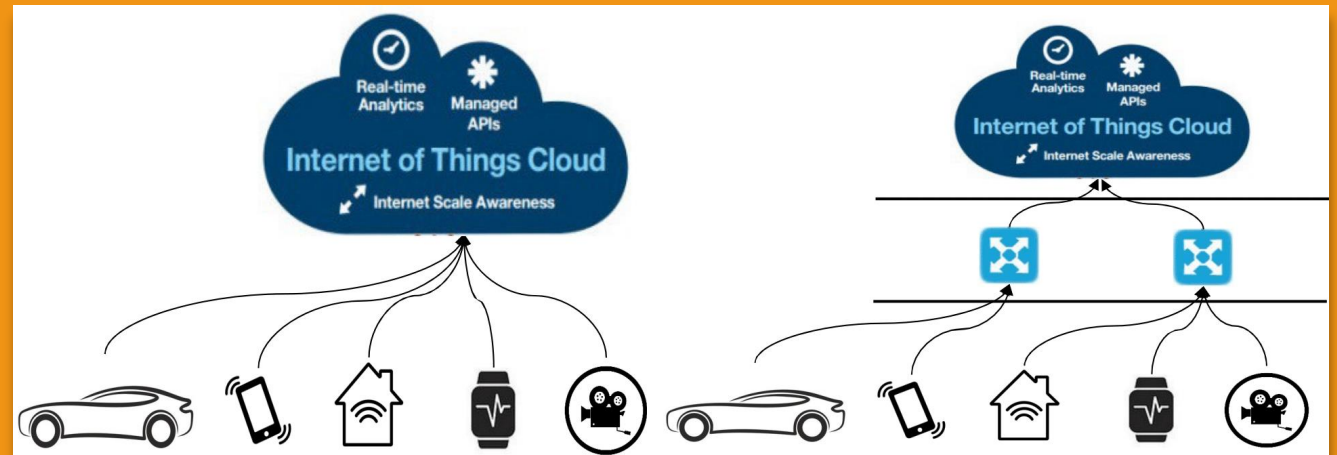
Distributed computing - Transparency

Transparency	Description
Access	Hide differences in data representation and how an object is accessed
Location	Hide where an object is located
Relocation	Hide that an object may be moved to another location while <i>in use</i>
Migration	Hide that an object may move to another location
Replication	Hide that an object is replicated
Concurrency	Hide that an object may be shared by several independent users
Failure	Hide the failure and recovery of an object

Internet of Things (IoT)



IoT Cloud and Edge



The three big players

- AWS: Amazon Web Services ~18 y/o
- Microsoft Azure ~16 y/o
- Google Cloud Platform ~16 y/o

aws



Azure



Google Cloud Platform

What does it mean for programming?

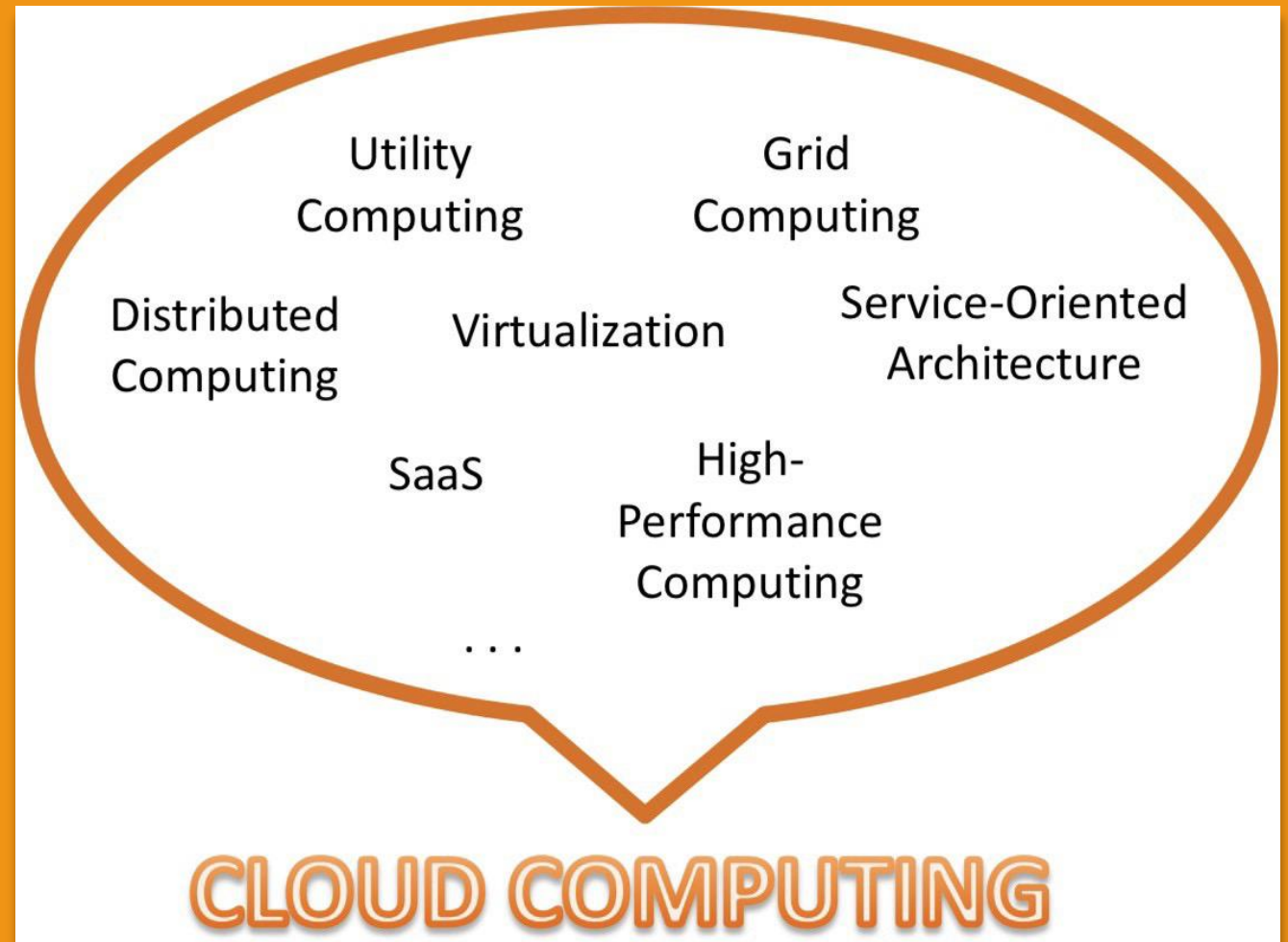
- System programmers/admins/devops
 - Building and deploying cloud systems and services
 - Focus: Virtualization, software configuration management, programming, scripting (e.g., C, bash)
- Programmer
 - Using cloud services
 - Focus: Development, deployment, etc.

What does it mean for programming?

- Cloud programming is about
 - Using vendor APIs
 - Developing our own services (and APIs)

Take aways

- Cloud computing is a paradigm in which information is permanently stored in servers on the internet.
- The future for cloud computing is bright. The big companies are investing lots of resources into this.



Take aways

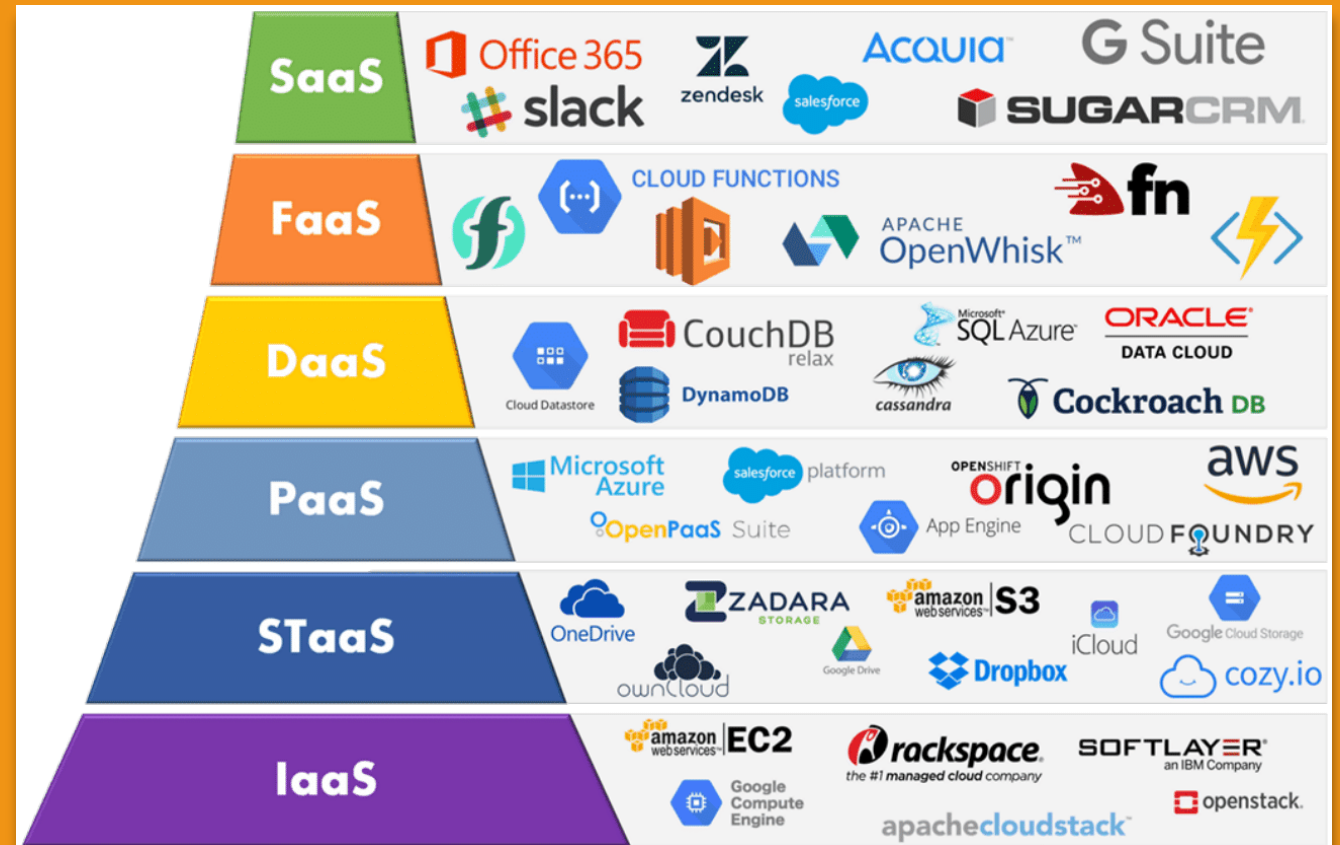
Advantages

- Increased reliability
- Device independency
- Unlimited storage
- Remote access
- Eco-friendly*
- Economical*

Disadvantages

- Totally internet connection oriented
- Require high bandwidth of net connection
- Vendor dependant for data maintenance

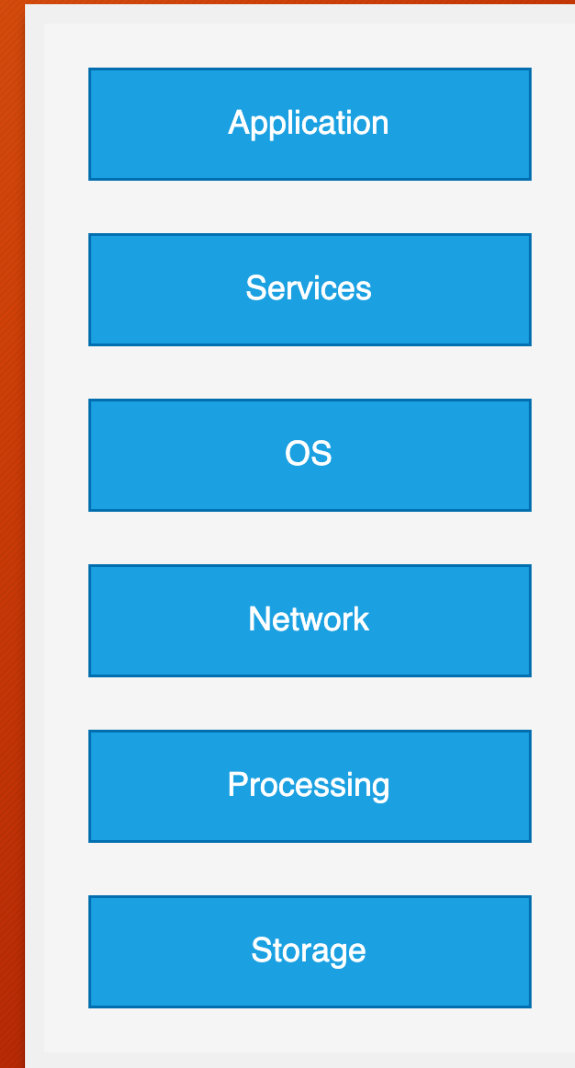
Cloud computing stack



Cloud computing stack

Infrastructure as a Service

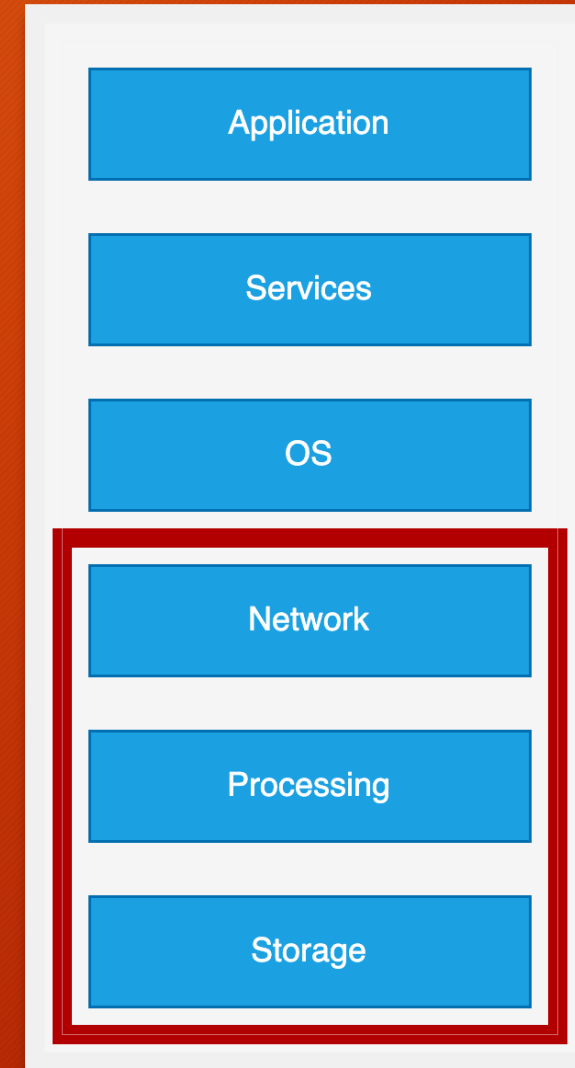
What is managed for you?



Cloud computing stack

Infrastructure as a Service

What is managed for you?



IaaS examples

- Amazon Web Services
- Google Cloud Platform



Google Cloud Platform

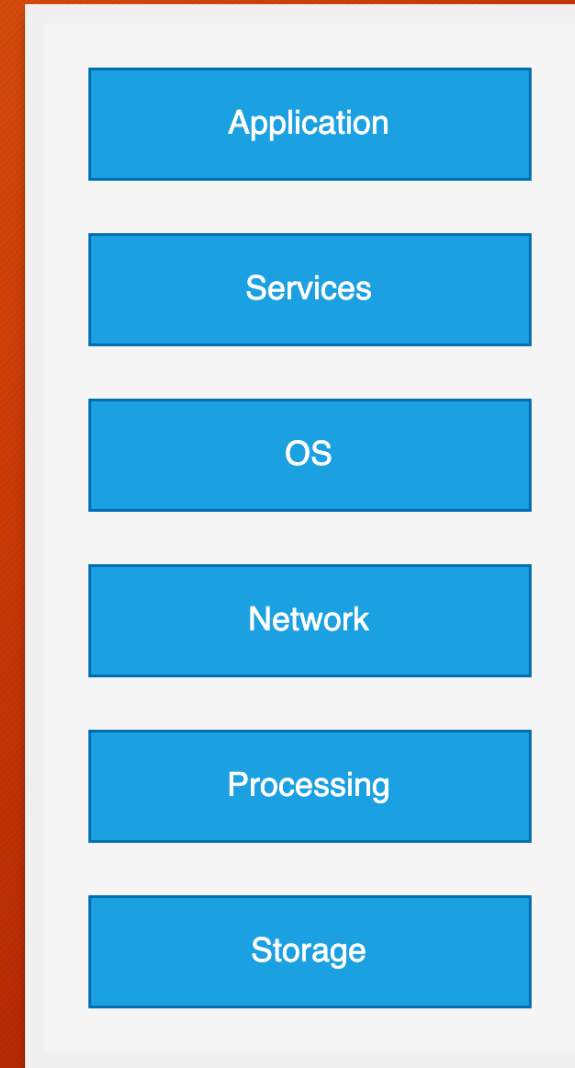


amazon
web services™

Cloud computing stack

Platform as a Service

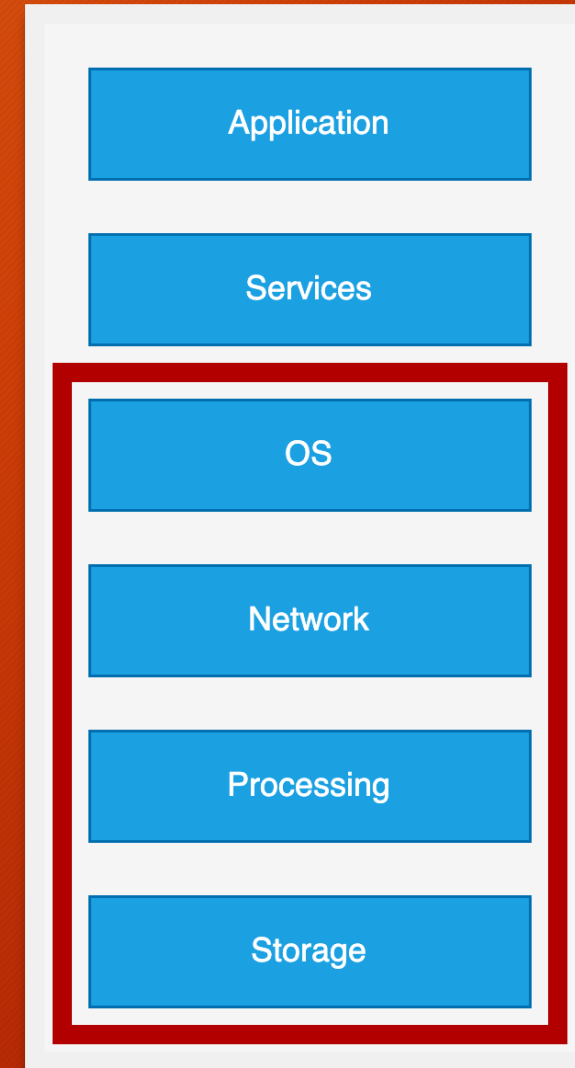
What is managed for you?



Cloud computing stack

Platform as a Service

What is managed for you?



PaaS examples

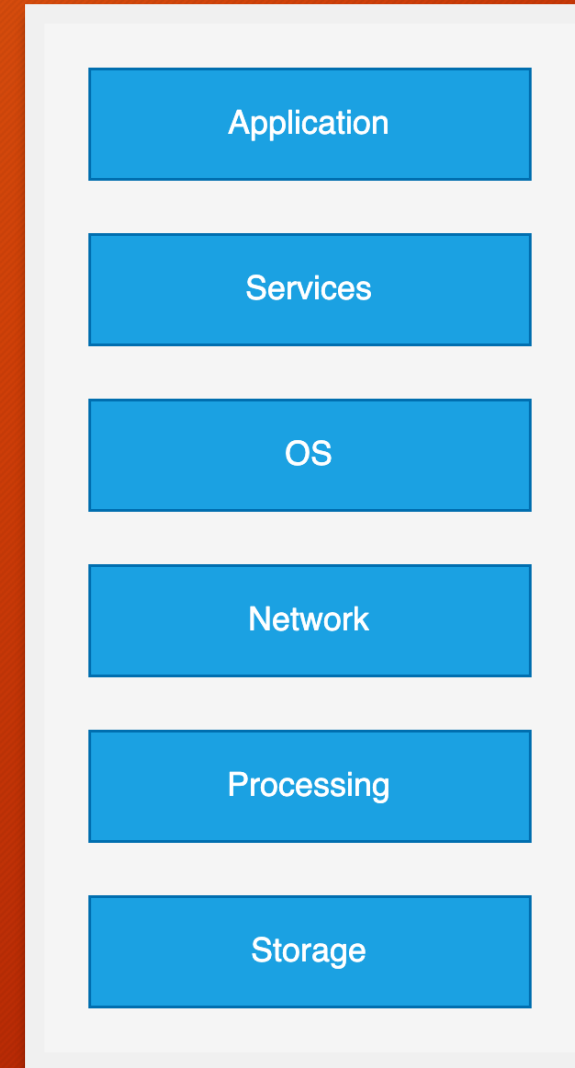


heroku

Cloud computing stack

Software as a Service

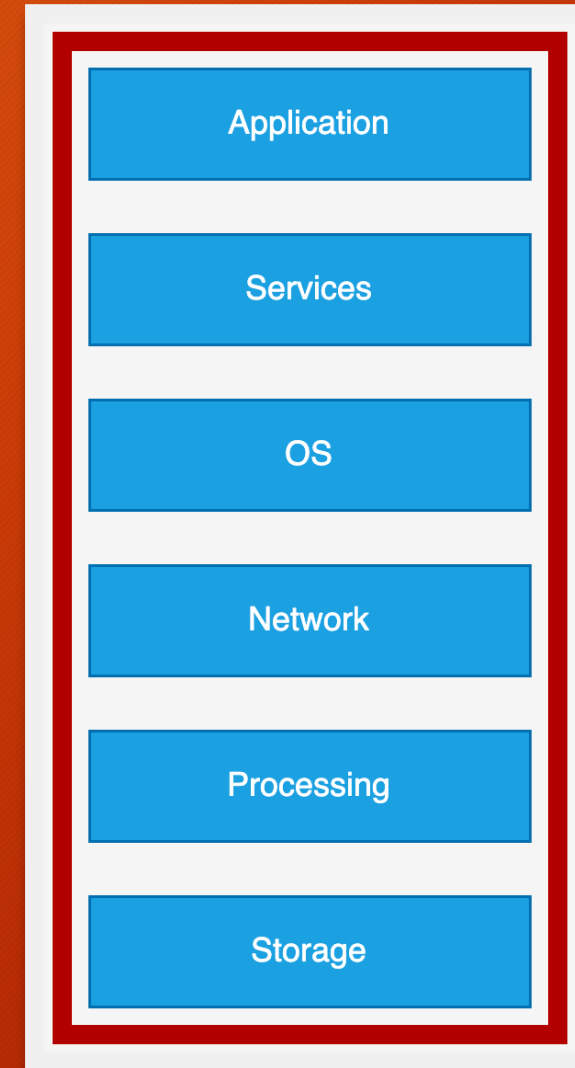
What is managed for you?



Cloud computing stack

Software as a Service

What is managed for you?



SaaS examples



Yey :D

- Send me an “about me” mail, so I get to know you :D
- paul.knutson@ntnu.no